Software Requirements Specification

(123) - Flight Reservation System

TA: Radwa Moustafa

Members:

Name	ID	Department
سارة طارق محمدعبدالحليم	2023170251	SC
هاجر علي محمد محمود علي	2023170684	SC
مرام محمد السيد احمد طه	2023170568	SC
سارة سعيد محمود عبد الجواد علي	2023170250	SC
محمد صادق يوسف علي	2023170504	SC
ميرفت هشام يحي محمود	2023170653	SC

1. Introduction

The Flight Reservation System is designed to facilitate the end-to-end process of searching, booking, canceling and managing airline flight tickets. The system serves both travelers and airline administrators by providing a centralized, reliable, and user-friendly environment for all flight-related operations.

The Flight Reservation System connects to an airline database to manage flight schedules, seat availability, and pricing. It also interacts with a payment module to process booking transactions. Additionally, the system connects with an email notification service to send booking confirmations and reminders to users.

This system aligns with the strategic objective of modernizing airline booking operations, reducing manual overhead at booking counters, and improving customer satisfaction through a seamless digital experience.

2. User Requirements

As a Passenger, I want to:

- Register a new account by providing personal details (name, email, password, phone number, and date of birth).
- Log in to the system using email and password to access booking features.
- Search for available flights by specifying origin, destination, travel date, and number of passengers.
- View detailed information about each available flight including airline, departure/arrival times, duration, stops, and price.
- Book a flight by providing passenger details and selecting a seat.
- Pay for the booking using the available payment methods within the system.
- Receive a booking confirmation displaying the e-ticket and PNR (Passenger Name Record) number.
- Cancel or modify my reservation before departure and receive refund.
- View my booking history and current reservations in my personal profile.
- Check in through the system within 24 hours before departure.

As an Airline Administrator, I want to:

- Log in to the system using email and password to access administrator features.
- Add, update, and remove flight schedules and seat availability.
- Search for a specific passenger's reservation by name or PNR.
- Manage passenger check-in status for each flight.

- View all reservations made for each flight.
- View reports on seat occupancy and revenue for each flight.
- Manage customer accounts and handle special service requests (e.g., meal preferences).
- Send notifications to passengers about flight delays, cancellations, or gate changes.

3. Functional Requirements

Functional Requirement 1: User Registration and Login

Description/Action: The system allows users to create an account and log in securely.
Requirements/Inputs: Name, email, password, phone number, date of birth.
Source: User input through registration/login forms.
Pre-condition: User is not already logged in (for registration).
Post-condition: User account is created or user is authenticated.
Output: Confirmation message or successful login access.

Functional Requirement 2: Search for Available Flights

Description/Action: The system allows the user to search for available flights based on specific criteria.
Requirements/Inputs: Departure city, arrival city, travel date, number of passengers.
Source: User input through the search interface.
Pre-condition: The system is connected to the flight database.
Post-condition: A list of matching flights is displayed.
Output: List of flights including flight details (time, price, duration, available seats).

Functional Requirement 3: Book Flight and Seat Selection

Description / Action: The system shall allow users to book flights and select seats.
Requirements / Inputs: Flight ID, passenger details, seat selection.
Source: User (Passenger).
Pre-condition: User is logged in and seats are available.
Post-condition: Reservation is created.
Output: Booking summary with total price.

Functional Requirement 4: Process Payment

Description/Action: The system shall allow users to complete payment for their booking through the available payment methods within the system.
Requirements/Inputs: Booking ID, payment method, payment details.
Source: User input.
Pre-condition: Reservation exists and the user is logged in.
Post-condition: Booking status is updated to "Confirmed" after successful payment.
Output: Payment confirmation and receipt displaying the booking ID, amount paid, and payment method.

Functional Requirement 5: Issue E-Ticket

Description/Action: The system generates and displays an electronic ticket after payment is completed.
Requirements/Inputs: Confirmed booking ID.
Source: System database.
Pre-condition: Payment is completed.
Post-condition: E-ticket is generated and stored.
Output: PDF ticket with flight details.

Functional Requirement 6: View Booking History

Description/Action: The system allows users to view their current and past bookings and request a cancellation.
Requirements/Inputs: Booking ID, user account credentials.
Source: System database.
Pre-condition: User is logged in.
Post-condition: Booking information is displayed or updated.
Output: List of bookings and updated status.

Functional Requirement 7: Online Check-In

Description/Action: The system shall allow passengers to check in for their upcoming flight through the system within 24 hours before the scheduled departure time.
Requirements/Inputs: PNR number or booking ID.
Source: Passenger input through the check-in interface.
Pre-condition: Passenger is logged in, has a confirmed booking, and the check-in window is open (within 24 hours before departure).
Post-condition: Check-in is recorded in the system, and the booking status is updated to "Checked In."
Output: A boarding pass displaying the passenger's name, flight details, seat number, and gate information.

Functional Requirement 8: Passenger Search

Description/Action: The system shall allow the administrator to search for a specific passenger's reservation by name or PNR number and display their full booking details.
Requirements/Inputs: Passenger name or PNR number.
Source: Admin input through the reservation search interface.
Pre-condition: Admin is logged in and booking records exist in the system.
Post-condition: The passenger's reservation details are retrieved and displayed.
Output: Full booking details including passenger name, flight information, seat number, booking status, and payment status.

Functional Requirement 9: Generate Reports

Description/Action: The system shall allow the administrator to generate reports on flight performance including seat occupancy and revenue.
Requirements/Inputs: Flight ID or date range selected by the admin.
Source: System database.
Pre-condition: Admin is logged in and booking data exists in the system.
Post-condition: Report is generated and displayed.
Output: Report showing total bookings, seat occupancy rate, and revenue per flight.

Functional Requirement 10: Send Notifications

Description/Action: The system shall allow the administrator to send notifications to passengers regarding flight delays, cancellations, or gate changes.
Requirements/Inputs: Flight ID, notification message, affected passengers list.
Source: Admin input.
Pre-condition: Admin is logged in and the affected flight exists in the system.
Post-condition: Notification is recorded in the system and delivered to affected passengers.
Output: Confirmation that the notification was sent successfully.

4. Non-Functional Requirements

Non-Functional Requirement 1: Performance

The system shall load and display flight search results within 3 seconds under normal operating conditions. Booking and payment processing shall be completed and confirmed within 5 seconds of user submission.

Non-Functional Requirement 2: Reliability

The system shall handle unexpected errors gracefully without crashing. In case of a database connection failure, the system shall display a clear error message to the user and maintain data integrity by rolling back any incomplete transactions.

Non-Functional Requirement 3: Security

All user passwords shall be stored in the database using encryption. The system shall enforce role-based access control, ensuring passengers and administrators can only access features permitted to their role. Admin accounts shall only be accessible through verified admin credentials.

Non-Functional Requirement 4: Usability

The system interface should be simple and intuitive enough for a new user to complete a flight booking within 5 minutes without prior training.

Non-Functional Requirement 5: Scalability

The system shall be designed to support an increasing number of flight records, passenger accounts, and booking transactions without requiring major changes to the system architecture.

Non-Functional Requirement 6: Data Integrity

The system shall ensure that no duplicate bookings are created for the same seat on the same flight. All database transactions shall follow ACID properties (Atomicity, Consistency, Isolation, Durability) to guarantee that data remains accurate and consistent at all times.